

MAHMOUD ELLOUMI

Computer Engineering Student • Generative AI & AI Agents • Seeking a final-year internship (PFE)

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Computer engineering student (ENIS) who has integrated LLMs and generative AI into full-stack applications (Ollama, Mistral AI, n8n). Seeking a **final-year internship (PFE)** to design **AI agents** that solve real business needs.

PROFESSIONAL EXPERIENCE

Full-Stack & AI Developer Intern | IDSOFT (ID Software Solutions), Sfax

07/2026 – 08/2026

AI-enhanced after-sales service management app • React, Spring Boot, PostgreSQL, Ollama

- Built an after-sales web app (7-state service-request lifecycle) for 3 roles: customer, technician, manager.
- Integrated 6 local generative AI features (Ollama, no data sent to third parties): troubleshooting assistant, natural-language data assistant, request classification, best-technician suggestion, report-writing assistance.
- Designed a layered Spring Boot REST API (Swagger), secured with JWT, TOTP, BCrypt and role-based access control.

Mobile Developer Intern | LOGICOM Informatique, Sfax

07/2025 – 08/2025

Product scanning mobile app • Flutter

- Built a Flutter mobile app that scans a product's QR code to display its available prices and product details.
- Connected the app to the company database to retrieve prices and records for 40 products.
- Delivered 3 screens: home, scanner and cart.

PROJECTS

SPARKY – Intelligent Business Document Processing (PFA)

05/2026

Team of 3, Scrum (9 sprints) • Angular, FastAPI, Mistral AI, PostgreSQL, n8n

- Designed the n8n workflow (11 nodes, Google Sheets, email) and the Angular 20 frontend: scanner, price comparison, dashboard.
- Extracted invoices, quotes and purchase orders with Mistral AI in 3–8 s each, vs 120–180 s for our n8n + Ollama prototype.
- Compared 3 suppliers across 10 items in under 1 minute, vs about 20 minutes by hand.

Intelligent Traffic Monitoring – individual course project

2025

- Detected and classified vehicles in real time on video streams with YOLO and OpenCV (Python).
- Built a traffic analytics dashboard: vehicle counts and breakdown by vehicle type.

TERRANAUT – Drone-Based Detection of Sick Animals – team course project

2024

- Applied computer vision and deep learning (object detection) to thermal images captured by drones.

EDUCATION

Engineering Degree in Computer Science and Applied Mathematics | National Engineering School of Sfax (ENIS)

2024 – Present

Preparatory Cycle in Technology • top 100 in the national ranking | IPEIS, Sfax

2022 – 2024

Baccalaureate in Technical Sciences | 15 November 1955 High School, Sfax

2022

TECHNICAL SKILLS

AI & LLMs: LLMs (Ollama, Mistral AI, Qwen), Generative AI, AI agents, Machine Learning

Vision & Automation: Deep Learning, OpenCV, YOLO, PyTorch, OCR, n8n

Languages: Python, Java, JavaScript/TypeScript, C

Web & Mobile: React, Angular, Spring Boot, FastAPI, Node.js, Flutter

Data & DevOps: PostgreSQL, MySQL, MongoDB, Kafka, Docker, Git, Linux, Scrum

CERTIFICATIONS

NVIDIA Deep Learning • Cisco C Advanced • Scrum (SFC) • Hedera (HCF) • Microsoft MS-900

VOLUNTEERING (IEEE)

Photographer – IEEE ENIS SB (2026) • **Treasurer & Webmaster** – IEEE Cyber Security ENIS (2025)

Social Media Manager & Photographer – TSYP 13 • **Organizer** – Pitch Yourself for a Career

LANGUAGES & INTERESTS

Arabic: native • **French:** upper-intermediate • **English:** intermediate • **Spanish:** elementary

Interests: photography, art, sports, innovation